

Project

Guidelines and deadline

You are asked to form groups of **2/3 students max.**

Each group has to chose one of the following problems (see next slides), perform experimental analys as indicated, and write a report including:

1. a description of the adopted solution **4 points**
2. designed algorithms plus related global comments/description **4 points**; comments to main fragments of code **3 points**
3. experimental analysis, concerning in particular scalability **3 points**
4. comments about the experimental analysis outlining weak and strong points of the algorithms. **3 points**
5. an appendix including all the code. **2 points**

A pdf version of the report has to be sent via email by **April 15, 2026.**

Finding connected components in graph

- The algorithm is described in this paper
 - <https://www.cse.unr.edu/~hkardes/pdfs/ccf.pdf>
- The work to consists of understanding the MapReduce algorithm, and coding it into Spark by using both RDD and DataFrames
- Both Python and Scala implementations must be provided
- Experimental analysis comparing the RDD and DataFrame versions has to be conducted on graphs of increasing size

Data preparation for US flight delay prediction

- Consider the following paper
 - <https://www.dropbox.com/s/4rqnjueuqi5e0uo/TIST-Flight-Delay-final.pdf>
- Half of the paper is dedicated to data preparation by preprocessing and opportunely *joining* complex datasets about flights and weather conditions
- CSV files are here <https://www.dropbox.com/sh/iasq7frk6f58ptq/AAAzSmk6cusSNfqYNYsnLGIXa>
- The project consists in porting in Spark (DataFrame) the data preparation process.
 - Scala or Python
 - **RDD and DataFrames for the data wrangling part (the optional ML part only in DataFrames)**
- The report should detail each step, comment encountered difficulties and how these have been overcome.
- Optional: the group can then opt for performing in SparkMLinb the prediction analysis by using decision trees.
- Suggestion: a group of three students would be preferable, dataprep is time consuming.
- Why not: once data is prepared use Spark ML for prediction, by using ML techniques adopted in the paper, or other ones you deem more efficient.

Fast Matrix Factorization for Online Recommendation with Implicit Feedback*

- Consider this paper
 - <https://www.dropbox.com/s/1nqw7zvmo91gq3a/IMPORTANT-70805024%20copy.pdf>
- It describes/presents an efficient matrix factorization ALS algorithm
- The paper shows that the algorithm can be easily parallelized, but no MR implementation is provided
- The work consists of finding a Python Spark version of the algorithm (**RDD OR DataFrames**) and perform experimental analysis in one of the dataset used by the paper.

Streaming Kmeans*

- This project is based on this Master thesis
 - https://www.dropbox.com/s/50rlhd5a2dc5s4b/Bitsakis_Theodoros_Dip_2018%20copy.pdf
- It describes a streaming algorithm for Kmeans, which is implemented in Flink.
- In the project you are required to provide a Spark Streaming implementation
- Ideally use Scala, but Python is good too.
- Write a technical report, around 20 pages max, with explanations of the main points about your implementation and experiments.

K-NN or Naive Bayes

- K-NN
 - <https://www.dropbox.com/scl/fi/3wwph2l0b8a2k1scrvclq/A-MapReduce-based-k-Nearest-Neighbor.pdf>
- Naive Bayes
 - <https://www.dropbox.com/scl/fi/04pv23898iqxukbib3doy/Na-ve-Bayes-Classifier-A-MapReduce-Approach.pdf>
- The work to consists of understanding the MapReduce algorithm, and coding it into Spark by using both RDD and DataFrames- just Python
- Experimental analysis, use datasets in the paper or other of you choice, and reduce the size eventually if needed .

You can mount your own project

- Find an interesting problem admitting MapReduce implementation in Spark
- Write a one-page proposal and submit it to me (by email).
- I will let you know whether the proposal is good and eventually how to adapt it.
- RDD and/or DataFrame implementation, Python and/or Scala, depending on the project.